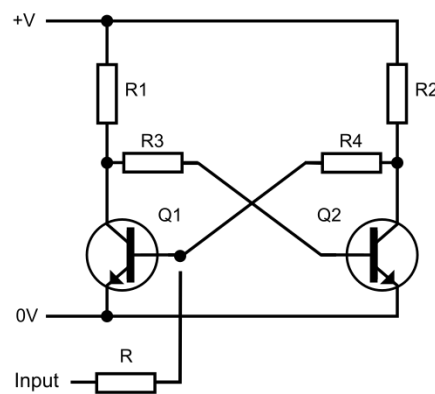


ENGR2420 Final Project Proposal

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For our final project, we want to implement a BJT schmitt trigger collector-base coupled circuit on a breadboard with 2N23904 BJTs. We want to observe the hysteresis behavior of the circuit by measuring the upper and lower threshold voltages. We will use an oscilloscope to get the input/output transfer characteristic and show the circuit's ability to convert a sine wave into a square wave. We will then build the circuit in LTspice to demonstrate how the values of the coupling resistors impact the hysteresis response.



BJT bistable collector-base coupled circuit converted to a Schmitt trigger by connecting an additional base resistor to one of the bases. Source: [Wikipedia](#)

Resources:

- https://en.wikipedia.org/wiki/Schmitt_trigger#:~:text=It%20is%20an%20active%20circuit,sufficiently%20to%20trigger%20a%20change.
- <https://studylib.net/doc/8150186/schmitt-trigger---wikipedia--the-free-encyclopedia?p=4>
- <https://www.scribd.com/document/259067402/Schmitt-Trigger>
- <https://www.edn.com/schmitt-trigger/>
- <https://www.electronics-tutorials.ws/amplifier/common-collector-amplifier.html>
- <https://www.digi-electronics.com/en/blogs/schmitt-trigger-function-hysteresis-circuit-design-and-applications/210.html>

Articles to gain context (and for funsies):

- <https://ieeexplore.ieee.org/document/10849181>
- <https://ieeexplore.ieee.org/document/9827203>
- <https://ieeexplore.ieee.org/document/10084272>
- <https://ieeexplore.ieee.org/document/1620839>